



Evaluating Depth Fusion for Instance Segmentation in Fruit Harvesting Robots

Edward John Domingo (22164775)

Project report for Postgraduate Project B

Supervisor: Yi Guo

*School of Computer, Data and Mathematical Sciences,
Western Sydney University*

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Abstract

Reliable visual perception is essential for fruit harvesting robots, which must detect and delineate individual fruits under occlusion and varying conditions. Depth is a readily available complementary signal that can help address these colour-based limitations. This study evaluates whether the augmentation of depth would improve transformer-based instance segmentation model performance. An RGB baseline is assessed against different RGB-D variants including an early fusion via a widened patch-embedding convolution and an iterative cross-attention mechanism. All variants are fine-tuned from identical COCO-pretrained weights. Contrary to our hypothesis, no fusion variant improved over the RGB baseline, which scored highest on every metric. A patch-embedding analysis shows depth exerts minimal or misdirected influence.

Keywords: RGB-D fusion, instance segmentation, automated fruit harvesting

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1 Introduction

Fruit production is a major component of global agriculture. However, fruit harvesting remains a highly labour-intensive process, requiring skilled workers to manually identify, select and harvest ripe fruits. Given that this work is often seasonal and situated in rural areas, a major challenge in fruit production and harvesting is its vulnerability to labour shortages. This leaves a gap between the global demand for fruits and the availability of a workforce to harvest them. Robotic harvesting has emerged as a potential solution, reducing dependence on seasonal labour and standardising harvesting quality (Zhou et al., 2022).

A requirement for any harvesting robot is a reliable visual perception module. Before the fruit may be picked, the system must first detect and delineate individual fruits. Perception in the field contends with variable lighting, heavy occlusion within dense fruit clusters and colour ambiguity between fruits and foliage. Harvesting robots are typically also equipped with depth sensors to use for 3D localisation, making depth available as a complementary signal for these perception challenges. This project therefore investigates whether adding depth improves fruit instance segmentation through a controlled comparison of fusion strategies against a colour-only baseline.

2 Background and Literature Review

2.1 Related Work

2.1.1 Instance Segmentation

Over the past decade, deep learning methods have replaced traditional hand-crafted features for fruit perception (Zhou et al., 2022). Examples include the use of Mask R-CNN and YOLO-based segmentation for apple orchards (Sapkota et al., 2024), and more recent transformer-based backbones, which have been applied to tomato instance segmentation (Shinoda et al., 2023). Across the literature, two failure modes are identified: occlusion within fruit clusters and colour ambiguity of green fruit against green foliage. (Bargoti and Underwood, 2017; Sapkota et al., 2024). Both are limitations of colour-only input, which motivates the augmentation of depth to the standard colour channels.

2.1.2 Depth and RGB-D Fusion

The addition of depth is expected to improve perception in cluttered scenes where colour alone is lacking. Various studies have been done to add in depth for agricultural use cases and differ on fusion strategies. Gené-Mola et al. (2019)

adapted the input layer of a Faster R-CNN model to take in a five-channel RGB-D intensity input for Fuji apple detection. On the other hand, Tu et al. (2020) trained independent RGB and depth detectors and fused them at the decision level for passion fruit detection. However, direct comparison across fusion stages against a colour-only baseline remains relatively limited, particularly for instance segmentation using transformer architectures.

2.2 Theoretical Foundations

2.2.1 SwinTransformer and Mask2Former

The following are the key architectures implemented in this project:

The Swin Transformer serves as the backbone feature extractor. It produces hierarchical, multi-scale feature maps using windowed self-attention. It yields a CNN-like feature pyramid that is suited to the multi-scale nature of instance segmentation (Liu et al., 2021).

Mask2Former is an instance segmentation model and builds on this backbone with a pixel decoder and a transformer decoder. A fixed set of N learnable queries is refined into candidate instances. Each produces a class label and a mask embedding that combines with the per-pixel embedding to yield a binary mask per query (Cheng et al., 2022).

Training uses bipartite (Hungarian) matching to assign queries against ground-truth instances. It combines a classification term (cross-entropy) with a mask term (binary cross-entropy plus Dice loss). Inference outputs N pairs, each stored by class confidence and mask quality.

2.2.2 Evaluation Metrics

The key metric used to assess segmentation quality is Average Precision (AP), which builds upon the Intersection over Union (IoU) (Everingham et al., 2010; Lin et al., 2014).

The **Intersection over Union (IoU)** measures the spatial overlap between a prediction mask M_p and the ground truth mask M_g . It is defined as:

$$\text{IoU}(M_p, M_g) = \frac{|M_p \cap M_g|}{|M_p \cup M_g|}$$

and ranges from 0 (no overlap) to 1 (full overlap).

At a given IoU threshold, a prediction is classified as a true positive (TP) if the

predicted class is correct and the mask meets the threshold. Otherwise, it is counted as a false positive (FP). Unmatched ground truth instances are counted as false negatives (FN). Precision and Recall are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

Every prediction carries a confidence score in $[0, 1]$, with only detections scoring above the threshold retained. Raising the threshold increases precision but lowers recall and vice versa.

The **Average Precision (AP)** summarises this trade-off. Sweeping the confidence threshold produces the precision-recall curve, and AP is the area under this curve at a given IoU threshold. In multi-class applications, each class is evaluated independently with the results averaged across classes.

3 Hypothesis and Aim

This study seeks to evaluate whether augmenting standard colour information with depth improves instance segmentation performance for fruit harvesting applications. Given the higher capacity of the RGB-D models, we hypothesise that they will achieve higher segmentation metrics as compared to RGB-only models.

4 Methodology

4.1 System Design / Research Design

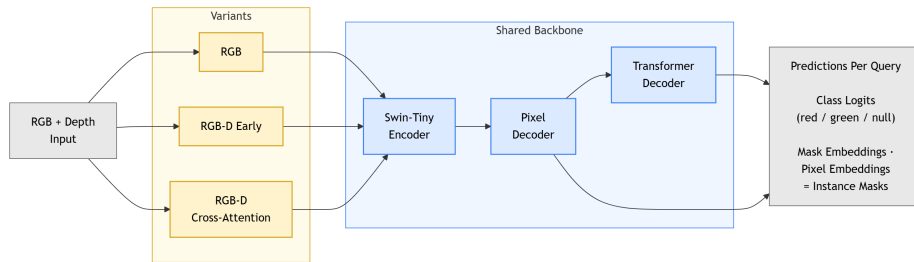


Figure 1: Architecture Diagram

The study centres on a controlled comparison between architectures. The base instance segmentation model uses a Mask2Former architecture with a Swin-Tiny backbone. Figure 1 shows the general architectural flow. Different variants are

slotted in the input stage that modifies the patch embeddings that are fed into the Swin-Tiny backbone. All other modules including the backbone stages, pixel decoder, transformer decoder and prediction heads are kept consistent across variants. All configuration choices such as optimiser, learning rate schedule, augmentations and data splits are kept consistent as well. Thus, the comparison isolates the effect of adding in depth via the initial patch embedding stages.

4.1.1 Variants

The variants that we use to show the effect of adding in depth are as follows:

RGB

This variant takes in the standard 3-channel RGB as input to the model. No changes has been made to the main architecture. This variant carries no depth information and will act as our baseline to compare against the depth-aware variants.

RGB-D Early Fusion

In this variant, depth is naively concatenated to the RGB image. The Swin backbone’s initial convolution is widened from $\text{Conv2d}(3, 96)$ to $\text{Conv2d}(4, 96)$ to account for the addition of the depth channel. The pretrained RGB weights are preserved and the new depth channel is zero-initialised.

RGB-D Cross Attention

RGB and depth are embedded by two separate patch-embedding modules that output token sequences with matching shapes. These are fused by a learned attention mechanism before the Swin-Transformer stages begin. An initial depth-to-RGB attention step is followed by K optional bidirectional refinement cycles. The value-projections of the attention blocks are zero-initialised. Only the refined RGB token stream is used as an input to the further Swin stages, with the depth stream discarded after fusion.

The mechanism is described by the equations below:

Let $S(\cdot) = \text{softmax}\left(\frac{\cdot}{\sqrt{C}}\right)$ (row-wise), with $X_D^{(0)} := X_D$.

Initial step (D guides RGB):

$$X_{RGB}^{(0)} = X_{RGB} + S\left((W_{D1}X_D)(W_{D2}X_D)^\top\right) W_{I1} X_{RGB}$$

For $k = 1, 2, \dots, K$:

$$X_D^{(k)} = X_D^{(k-1)} + S\left((W_{Iq}^{(k)}X_{RGB}^{(k-1)})(W_{Ik}^{(k)}X_{RGB}^{(k-1)})^\top\right) W_{Dv}^{(k)} X_D^{(k-1)}$$

$$X_{RGB}^{(k)} = X_{RGB}^{(k-1)} + S\left((W_{Dq}^{(k)} X_D^{(k)})(W_{Dk}^{(k)} X_D^{(k)})^\top\right) W_{Iv}^{(k)} X_{RGB}^{(k-1)}$$

Output: $X_{RGB}^{(K)}$ (the refined depth stream $X_D^{(K)}$ is discarded).

All variants are warm-started from the same COCO-pretrained weights. The RGB-D variants are constructed so that they are functionally equal to the RGB baseline at initialisation.

4.2 Implementation

4.2.1 Dataset

The dataset used in this study is Rob2Pheno, which was used in an earlier study by Afonso et al. (2020). It consists of RGB and depth image pairs (TIFF files) of tomato plants with size 720x1280. The depth maps are stored as 8-bit grayscale images (0-255). The dataset contains a total of 123 images that were manually annotated with two classes: redfruit (ripe) and greenfruit (unripe). Figure 2 show samples of the image pairs along with the annotated fruits. These were split into 83 training images (1074 instances of 365 red / 709 green) and 40 test images (538 instances of 176 red / 362 green).

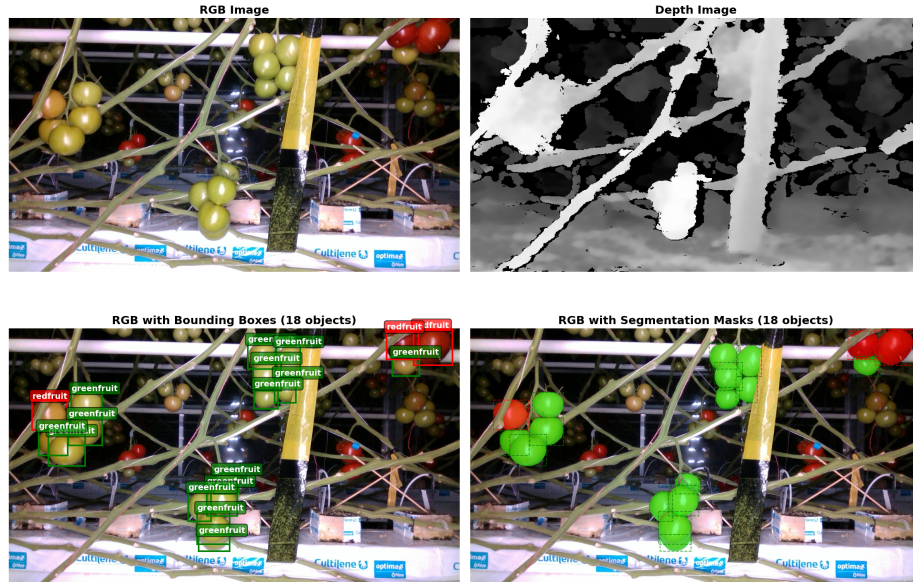


Figure 2: Sample RGB-Depth image pair with annotations from the Rob2Pheno dataset (Afonso et al., 2020)

4.2.2 Pre-processing and Augmentations

The Rob2Pheno depth TIFFs use a pixel value of 0 on locations where the sensor readings are invalid. Detectron2 normalises each input channel, thus, if left untreated, these invalid sensor readings leave a strong negative signal where there is no actual depth measurement. To mitigate this, we mean-fill the zeros. Invalid pixels are replaced with the training-split depth mean so that after normalisation they evaluate to exactly 0.0.

Augmentations were used to increase training images and are geometric in nature. This includes random horizontal flips and resizing. Colour-based augmentations such as colour jitter are omitted as these would corrupt the depth values.

4.2.3 Cross-validation Splits

A 5-fold cross-validation partition is generated once using a fixed seed and is reused across all variants. This ensures that every variant is trained and evaluated on the same fold assignments.

4.2.4 Training Configuration

Training involves fine-tuning from a COCO-pretrained model rather than training from scratch. Table 1 summarizes the hyper-parameters used for the final training run.

Table 1: Training hyper-parameters.

Hyperparameter	Value
Optimiser	AdamW, weight decay 0.05
Base LR / backbone mult.	1×10^{-4} / $0.1 \times$
Depth-path LR mult.	$5 \times$ (DCA only)
LR schedule	Cosine annealing to ~ 0 , no step decay
Warm-up	Linear, 200 iterations, start factor 0.01
Training budget	200 epochs
Batch size	2
Input resolution	Shortest edge 960–1280 (multi-scale)
Gradient clipping	L2 norm

The depth path in the DCA variants uses a 5x learning-rate multiplier, since its weights are zero-initialised and receive no information from the COCO-pretrained backbone. This allows the depth path to catch-up to the already

trained RGB path. No increase was applied to the RGBD Early variant where the depth channel is folded into the patch-embedding convolution and cannot be decoupled from the RGB channels.

4.3 Evaluation Plan

4.3.1 Metrics

Performance is measured with Detectron2’s COCOEvaluator after every 10 epochs on the in-fold validation splits. Table 2 defines the metrics collected

Table 2: Evaluation metrics.

Metric	Definition
AP	Average Precision over IoU thresholds 0.50 to 0.95 (step 0.05)
AP50	AP at a single IoU threshold of 0.50
AP75	AP at a single IoU threshold of 0.75
Per-class AP (redfruit, greenfruit)	AP averaged over IoU thresholds 0.50–0.95, computed separately for each class

4.3.2 Paired Statistical Comparison

Each CV fold uses an identical train/test split across variants, which means per-fold AP differences form matched pairs. We compare the RGB baseline against each fusion variant using per-fold differences ($\Delta\text{AP} = \text{AP}_{\text{RGB}} - \text{AP}_{\text{variant}}$). The mean difference, the number of folds in which RGB scored higher, and a paired two-sided t-test p-value are reported.

4.4 Hardware and Software

The Mask2Former model was built using the Detectron2 Python library (Wu et al., 2019). It uses PyTorch as its deep-learning framework. The full set of dependencies, the environment specifications and set-up are provided in the following repository: <https://github.com/emdomingo/tomato-fruit-instance-segmentation>. All training was performed on the Wolffe HPC cluster on the ampere40 and ampere80 partitions.

5 Results

5.1 Overall Segmentation Accuracy

A five-fold cross validation was done on the training set and were evaluated on AP, AP50, AP75 and per-class AP. Table 3 shows the mean and standard deviation across folds taken at the best epoch checkpoint of each split. Results for other training configurations can be found on Appendix A and follow the same trend.

Table 3: 5-fold cross-validation segmentation AP (mean $\pm$ std). Highest APs in bold.

Variant	AP	AP50	AP75
RGB	70.22 $\pm$ 1.84	94.93 $\pm$ 0.78	82.55 $\pm$ 1.18
RGB-D Early	67.37 $\pm$ 2.32	93.89 $\pm$ 2.33	78.67 $\pm$ 3.28
RGB-D DCA (K=0)	68.19 $\pm$ 2.12	94.15 $\pm$ 2.52	79.80 $\pm$ 1.66
RGB-D DCA (K=1)	67.43 $\pm$ 1.31	94.49 $\pm$ 1.58	77.95 $\pm$ 2.68
RGB-D DCA (K=2)	67.48 $\pm$ 2.02	93.90 $\pm$ 1.35	77.43 $\pm$ 1.93

Variant	AP (red)	AP (green)
RGB	70.84 $\pm$ 4.46	69.61 $\pm$ 2.89
RGB-D Early	68.63 $\pm$ 3.79	66.11 $\pm$ 3.75
RGB-D DCA (K=0)	69.06 $\pm$ 3.85	67.33 $\pm$ 2.81
RGB-D DCA (K=1)	68.46 $\pm$ 3.42	66.40 $\pm$ 2.54
RGB-D DCA (K=2)	68.02 $\pm$ 4.18	66.95 $\pm$ 3.68

Contrary to the hypothesis, the RGB-only variant performed best across all metrics, achieving 70.22 AP compared to the 67.37-68.19 AP for the RGB-D fusion variants. The gap is best seen in the stricter AP75 threshold, where RGB reaches 82.55 against the 77.43-79.80 of the RGB-D variants. At AP50, however, all variants are effectively tied, falling within 93.89-94.93 AP, with the spread smaller than the reported standard deviations.

Figure 3 shows validation AP across training for each variant. The RGB curve has an early advantage and remains above the curves associated for the fusion variants for AP, AP75 and both per-class metrics throughout training. In contrast, AP50 shows the variants converging into an overlapping band.

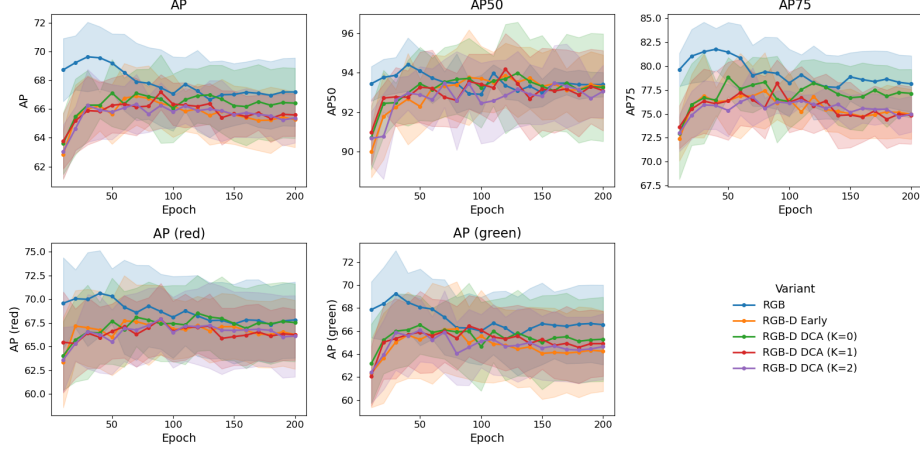


Figure 3: Validation AP across training for each variant, mean $\pm$ 1 std over the five cross-validation folds.

5.2 Paired Statistical Comparison

Table 4 reports the paired comparison of the RGB baseline against each fusion variant. RGB scored higher in all five folds, with the mean Δ AP falling between +2.03 and +2.85. The paired t-test rejects the null hypothesis of zero mean difference for all four comparisons ($p < 0.05$). However, with only five folds these tests are underpowered, and the folds are not fully independent since their training partitions overlap, so the p -values are treated as indicative.

Table 4: Paired comparison of RGB against each fusion variant across the five cross-validation folds. Δ AP is the per-fold AP difference (positive favours RGB).

Comparison	mean Δ AP	RGB wins	paired-t p
RGB vs RGB-D Early	+2.85	5/5	0.018
RGB vs RGB-D DCA (K=0)	+2.03	5/5	0.023
RGB vs RGB-D DCA (K=1)	+2.80	5/5	0.001
RGB vs RGB-D DCA (K=2)	+2.74	5/5	0.002

5.3 Visualisations

Inference was done on the fixed held-out test set which was excluded from every cross-validation fold during testing. For each variant, the median-performing fold (ranked by validation AP) was selected as the representative model. Figure 4 shows a sample of the instance predictions of every variant against the ground

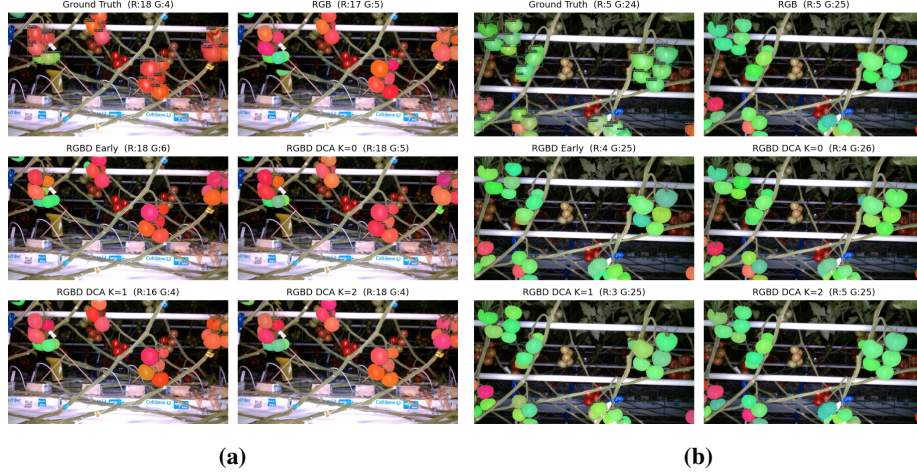


Figure 4: Sample visualisations on the test set using the median-performing fold. (a) Sample dominated by redfruit instances, (b) Sample dominated by greenfruit instances

truth. Inference against all test images is available in the git repository. A qualitative examination of the images show that all variants perform similarly. Individual images may favour one variant, but no variant shows a consistent advantage across the test set.

6 Discussion

6.1 Over-all Performance

Contrary to our hypothesis, none of the RGB-D fusion variants improved over the RGB-only baseline. RGB scored the highest on every metric (Table 3), with the gap clearest at the stricter AP75 threshold. All variants are practically tied at the more lenient AP50 threshold. The paired comparison (Table 4) shows RGB ahead in all five folds for every variant, and the qualitative inspection of the held-out test set (Figure 4) shows no variant with a consistent advantage. This means, that the model has not learned to use depth in a useful manner for the segmentation task.

Theoretically, the RGB-D variants should be able to converge towards the RGB baseline. Since the performance consistently falls below the RGB baseline, depth may be interfering with the RGB representation rather than being ignored.

6.2 Effects of Depth

To visualise the effect of depth, we compute the following per-patch relative depth contribution for the RGB-D variants. For the RGB-D Early, the patch-embed convolution is linear, that means the depth channel’s contribution can be isolated exactly. For the RGB-D Cross-Attention variants, we measure how much the depth-guided fusion step moves each RGB token off its initial RGB-only value.

$$\text{Early: } \frac{\| \text{conv}(\text{depth}) \|}{\| \text{conv}(\text{RGB}) \|} \quad \text{Cross-Attention: } \frac{\| X_{\text{fused}} - X_{\text{pre-fusion}} \|}{\| X_{\text{pre-fusion}} \|}$$

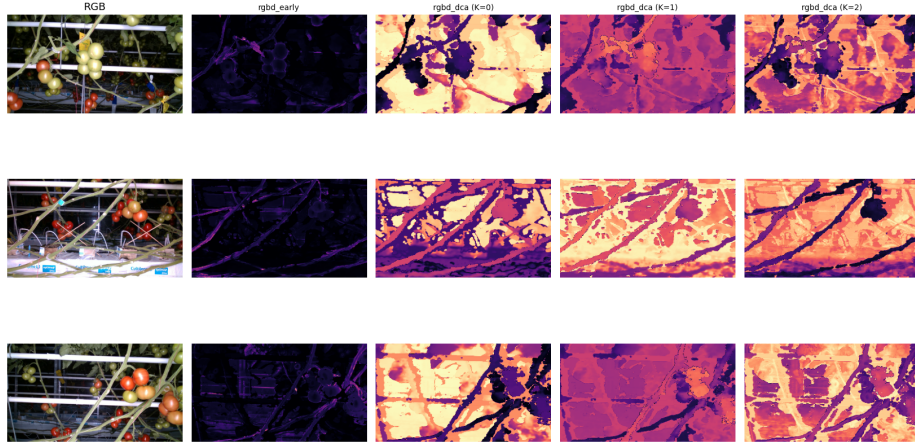


Figure 5: Patch Embedding Analysis (Colour Scale comparable within variants only)

This has been visualised in Figure 5 where brighter regions denote a stronger depth influence on the learned patch embeddings. The figure shows that depth has a minimal contribution on the patch embeddings for the RGB-D Early variant. This suggests that the model has converged without meaningfully exploiting the additional input.

For the DCA variants, the greatest influence that depth brings in the scene is on the background regions. Comparatively, there is little effect within fruit clusters. This is the opposite of what would have been useful where depth should ideally discriminate between overlapping fruits, but the model applies it on the region where it matters the least. This indicates that the models still rely predominantly on RGB for segmentation. This is a plausible explanation on why AP has not improved over the RGB baseline.

6.3 Limited Data

The training set only included 83 images, where transformers are data-hungry architectures. While this is semi-alleviated by fine-tuning on a pretrained model, this only affects the RGB side of the model as the pretrained backbone provides no prior information for the depth channels, whose weights have been zero-initiated and have no context yet. The RGB models converge quickly and we can assume that the RGB side of RGB-D variants does so as well. The depth side might need more training to catch-up, and in this process, the already good RGB representation could degrade.

6.4 Comparison with Other Studies

Afonso et al. (2020) implemented a Mask-RCNN model using the same greenhouse tomato dataset in this study. They used depth only as a post-processing filter by discarding predicted instances whose depth fell outside a stated foreground range. While a direct numerical comparison is not possible, as they reported precision, recall and F1 score, while we record AP, their findings with depth align with ours. Their filtering improved precision but lowered recall, this is attributed to errors in the depth sensor that discarded genuine foreground fruit. Our study carries out their recommendation to add in depth directly in the network and finds that it does not improve AP.

6.5 Future Recommendations

The addition of depth is implemented globally for all our RGB-D variants. However, the original failure modes we are trying to solve (clustering and difficulty distinguishing between green fruits and foliage), are inherently local problems. A cluster of fruits does not need the depth context of a separate cluster located in the same scene. This means the global fusion spends capacity relating regions that should remain independent.

Further more, the AP50 of every variant has scored at ~ 94 (Table 3) and the gap starts at the stricter AP75, where the quality of the mask matters more. This tells us that the coarse fruit detection is not the problem, but rather the refinement of the predicted masks.

Rather than fusing depth early in the backbone, a suggestion for a future architecture is to introduce depth after the mask predictions to refine the boundaries. This targets both the actual failure mode (mask precision at a high IoU threshold) and yields a less computationally-expensive architecture since depth will be processed for the local proposed regions rather than the full feature map.

7 Conclusion

This study evaluated whether augmenting RGB input with depth improves instance segmentation performance for fruit harvesting applications. As opposed to our hypothesis, no RGB-D fusion variant outperformed the RGB-only baseline, with RGB scoring the highest on every metric. This means the additional capacity of the RGB-D models did not produce better segmentation.

Analysing the patch-embeddings showed that depth exerts minimal influence on the learned representations for the RGB-D early variant, while for the RGB-D DCA variants, it exerts the strongest effect on the background rather than the overlapping fruit clusters where it would help the most. In addition, a small dataset and a non-pretrained depth pathway likely has left the depth branch undertrained as compared to the RGB branch.

All variants were comparable at AP50 (~ 94), with the performance gap emerging at the stricter AP75 threshold. This means the problem with the segmentation is in the mask refinement, rather than the coarse fruit detection. We therefore recommend future work to introduce depth later in the pipeline, as a stage after mask prediction, to refine the boundaries locally at a lower computational cost as opposed to the global application of depth.

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A Additional Results

The following section details results from other training set-ups.

Table A.5: 5-fold cross-validation segmentation AP (mean $\pm$ std) across training set-ups. Highest APs in bold (per set-up).

(a) Step schedule, full fine-tune					
Variant	AP	AP50	AP75	AP (red)	AP (green)
RGB	70.20 $\pm$ 1.90	95.17 $\pm$ 1.09	81.08 $\pm$ 2.43	71.01 $\pm$ 3.64	69.39 $\pm$ 2.60
RGB-D Early	67.88 $\pm$ 2.57	94.40 $\pm$ 0.85	77.88 $\pm$ 4.98	68.82 $\pm$ 4.76	66.94 $\pm$ 3.85
RGB-D DCA (K=0)	67.97 $\pm$ 2.02	94.50 $\pm$ 2.05	78.32 $\pm$ 3.22	68.98 $\pm$ 4.03	66.96 $\pm$ 3.19
RGB-D DCA (K=1)	66.91 $\pm$ 1.69	94.16 $\pm$ 2.22	77.36 $\pm$ 1.72	67.78 $\pm$ 3.91	66.04 $\pm$ 2.94
RGB-D DCA (K=2)	68.05 $\pm$ 1.30	94.05 $\pm$ 1.79	78.23 $\pm$ 1.26	69.61 $\pm$ 3.18	66.49 $\pm$ 2.81
(c) Cosine schedule, frozen backbone, dca-lr-mult 5					
Variant	AP	AP50	AP75	AP (red)	AP (green)
RGB	65.11 $\pm$ 2.56	90.51 $\pm$ 2.35	76.66 $\pm$ 2.15	66.11 $\pm$ 3.54	64.10 $\pm$ 2.47
RGB-D Early	56.86 $\pm$ 2.34	83.41 $\pm$ 3.57	66.56 $\pm$ 3.38	56.66 $\pm$ 3.67	57.05 $\pm$ 3.01
RGB-D DCA (K=0)	57.65 $\pm$ 2.97	83.62 $\pm$ 1.27	67.88 $\pm$ 3.68	58.70 $\pm$ 5.98	56.60 $\pm$ 2.88
RGB-D DCA (K=1)	57.82 $\pm$ 3.26	84.87 $\pm$ 3.36	67.50 $\pm$ 2.77	57.75 $\pm$ 3.32	57.90 $\pm$ 3.63
RGB-D DCA (K=2)	60.87 $\pm$ 1.23	87.27 $\pm$ 3.05	71.71 $\pm$ 3.43	61.49 $\pm$ 2.12	60.26 $\pm$ 3.63